

What Separates Winners?

A Data-Driven Analysis of NCAA Division I Men's Basketball

2025–26 Season · ESPN API · Snowflake + Python EDA

Target Audience: Athletic department strategy teams, conference offices, and basketball operations staff

362

D1 Teams

31

Conferences

11,954

Game Records

5,648

Players

The Central Question

What separates winning programs in NCAA D1 basketball — and can those factors be measured, quantified, and used to predict outcomes?

Our Approach — Five Areas of Analysis

1

The D1 Landscape

Map the structure of 31 conferences and 362 programs

2

Performance Drivers

Identify the metrics most strongly correlated with winning

3

AP Rankings Validation

Validate our metrics against the AP Top 25 external benchmark

4

Roster Composition

Examine whether physical and experience attributes predict success

5

Predictive Modeling

Build linear and logistic regression models to predict outcomes

Data Overview

All data from the ESPN Core and Site APIs for the 2025–26 season, processed through Snowflake into nine clean views.

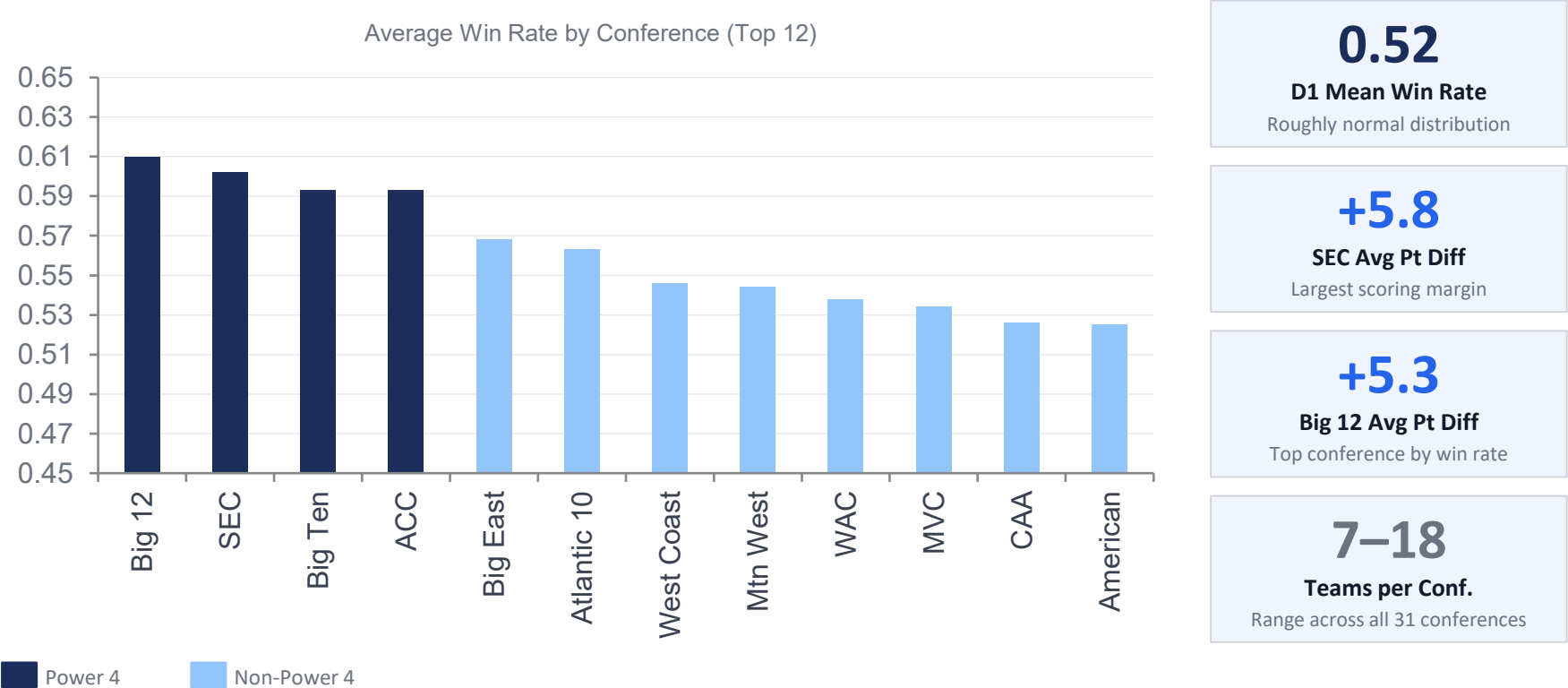


Table	Records	Description
Teams	362	Season-level stats: wins, losses, PPG, point differential
Games	11,954	Completed game results with scores and margins, one row per team per game
Unique Games	6,292	One row per game (deduplicated) — used for game-level scoring analysis
Players	5,648	Roster data: position, height, weight, years of experience

32 incomplete games removed (ESPN returned score = 0 with no win/loss result) · AP Top 25 poll merged as a second external validation dataset

The D1 Landscape

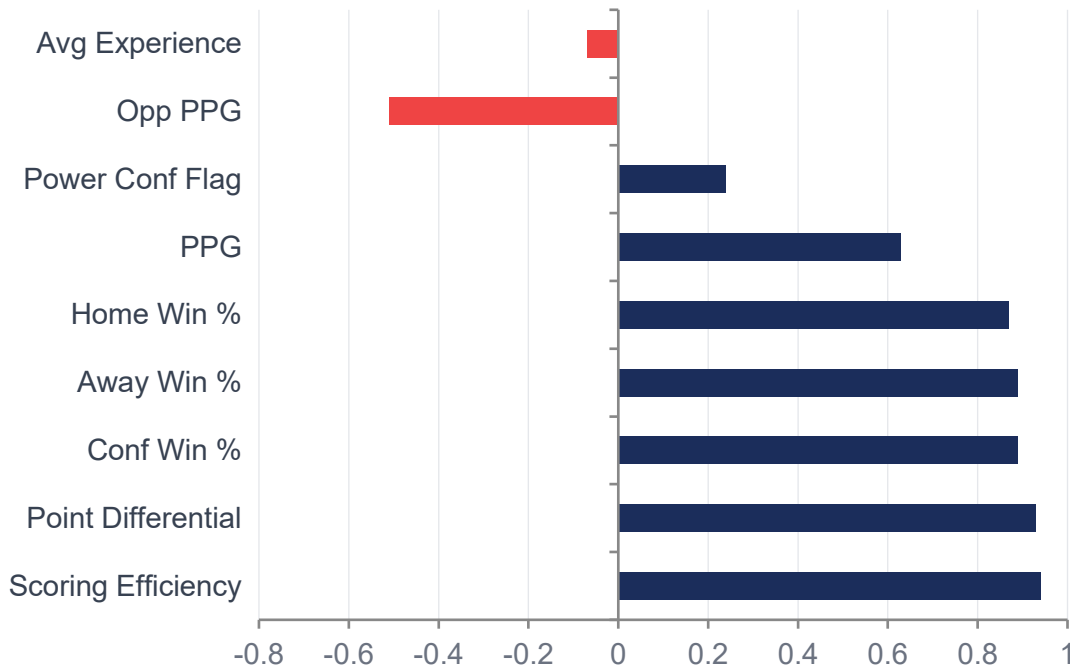
362 programs span 31 conferences — the four power conferences lead in both win rate and scoring margin, but the full distribution is wide.



What Drives Winning?

Raw scoring matters, but margin-based metrics and situational win rates align more closely with season success.

Feature Correlations with Win Rate

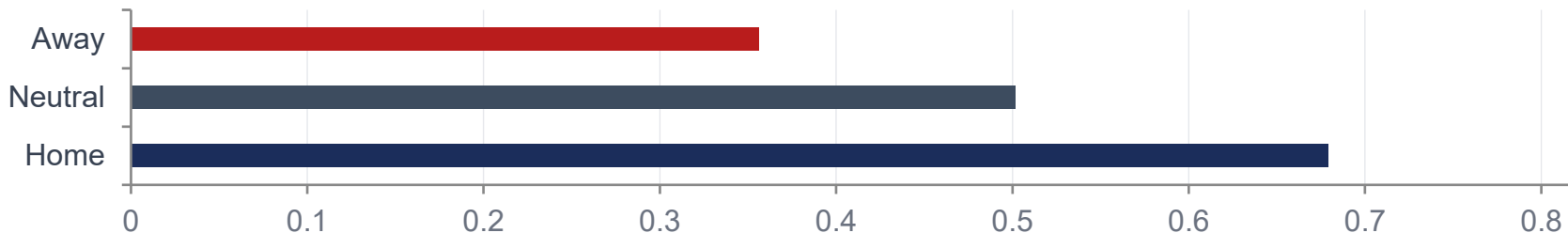
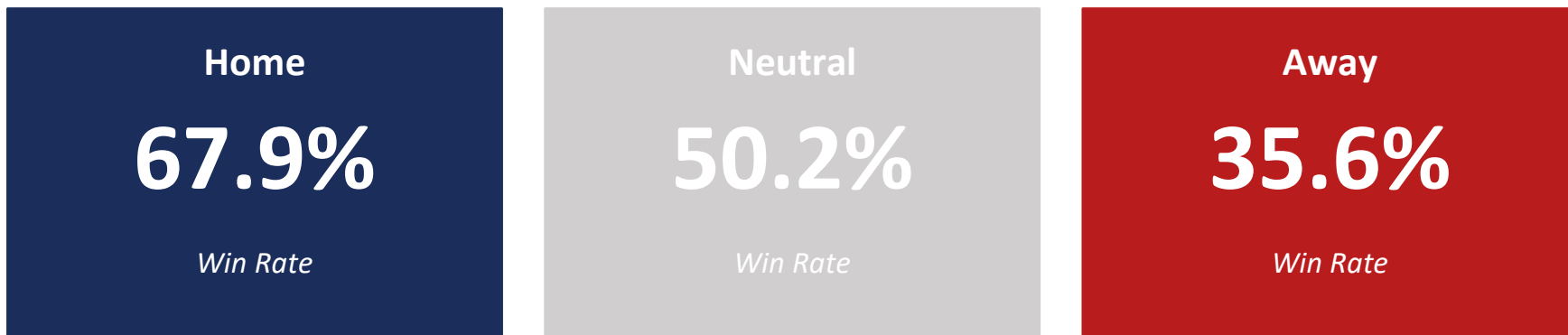


Interpretation

- Point differential correlates at 0.93 with win rate, making margin more informative than record alone.
- Scoring efficiency (PPG / Opp PPG) is the strongest single engineered metric at 0.94.
- Experience is close to zero (-0.07), suggesting average roster tenure alone is too blunt to explain success.
- Opp PPG is negative (-0.51): lower opponent scoring is associated with stronger records.

Home Court Advantage

Location alone provides a useful baseline before adding team quality features.



Neutral site games land at exactly 50.2% — when neither team has home court, the outcome is essentially a coin flip, which validates the data.

AP Rankings Validation

We merged the AP Top 25 poll from the ESPN API into our dataset to independently verify our metrics against expert consensus rankings.

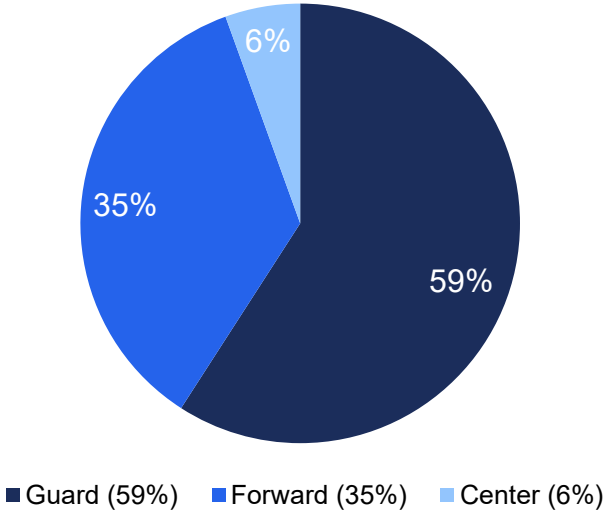
Metric	AP Ranked (25)	Unranked (337)	Gap
Win Rate	0.769	0.498	+27 percentage points
Points Per Game	82.05	75.20	+6.9 PPG more scored
Opp Points Per Game	70.34	74.28	-3.9 PPG fewer allowed
Point Differential	+11.72	+0.91	+10.8 pts per game gap

Key Finding: AP-ranked teams look meaningfully stronger across every core performance metric: they win more often, score more, allow fewer points, and post a +10.8 point differential advantage versus unranked teams. This validates point differential and scoring profile as useful indicators of program quality.

Roster Composition

Physical profile and experience show weak relationships with winning — D1 rosters are guard-heavy, physically consistent by position, and uniformly experienced.

Roster Composition by Position (5,635 players)



Physical Profile by Position

Position	Players	Avg Height	Avg Weight	Avg Exp
Guard	3,331	6'3"	190 lbs	2.55 yrs
Forward	1,993	6'8"	220 lbs	2.59 yrs
Center	311	6'11"	241 lbs	2.61 yrs

Key Finding: Roster experience is nearly unrelated to win rate. To evaluate roster quality, teams need richer data on production, recruiting, transfers, and minutes played.

Experience is uniform across positions (2.55–2.61 yrs) — roster turnover does not vary by position group. Top international countries: France (62), Nigeria (44), Senegal (41).

Predictive Modeling

Linear regression explains season win rate; logistic regression estimates game outcomes using location and season-level team quality features.

Model 1: Linear Regression

Target: Season Win Rate

R ²	0.978	97.8% of win rate variance explained
MAE	0.022	Avg error of 2.2 percentage points
RMSE	0.027	Root mean squared error

Top Model Signals

Home Win %

Away Win %

Scoring Efficiency

+0.398

+0.368

+0.112

Uses season-level performance features; best interpreted as explanatory

Model 2: Logistic Regression

Target: Individual Game Outcome (Win / Loss)

	Accuracy	ROC AUC	F1 (Win)
Location Only	64.9%	0.650	0.66
Location + Team Quality	68.1%	0.748	0.70

✓ Accuracy improves by 3.2 pts

✓ ROC AUC improves by +0.098

✓ Team quality adds signal beyond home court

Because team-quality features are season-level, this model is best interpreted as explanatory rather than a true pre-game forecast.

Key Takeaway: Location is a strong baseline predictor, but adding season-level team quality metrics improves model performance. The models reinforce the EDA: scoring efficiency and situational win rates are useful indicators of team strength.

Key Findings

1 Scoring margin is the clearest signal of team strength.

Point differential strongly correlates with win rate, while scoring efficiency is the strongest single correlated metric.

3 Home court advantage is real and measurable.

Home teams win 67.9% of games versus 35.6% for away teams, making location an important baseline factor in game outcomes.

5 Roster experience does not meaningfully predict winning.

Average roster experience is nearly uncorrelated with win rate, so evaluating roster quality requires richer data on production, recruiting, transfers, and minutes played.

2 Power 4 conferences perform better on average, but strong teams exist outside that group.

The Big 12, SEC, ACC, and Big Ten all exceed the D1 mean win rate of 0.52, while teams like High Point and Gonzaga rank among the national leaders in point differential.

4 AP rankings validate the core performance metrics.

AP-ranked teams average a +11.72 point differential versus +0.91 for unranked teams, suggesting the ESPN-derived metrics align with external expert rankings.

6 The models reinforce the EDA, but are best interpreted as explanatory.

The linear model explains season win rate well, while the logistic model improves from 64.9% to 68.1% accuracy when season-level team quality features are added to location.